Water Depth Inversion by Fusion of Active and Passive Remote Sensing Imagery With Waves Under the Super Learner–Neural ODE Framework

Aijun Cui, Yi Ma, Jingyu Zhang, and Ruifu Wang

***Abstract*—Most active and passive remote sensing imagery acquired over marine areas contains wave textures, presenting both challenges and opportunities for water depth retrieval. Although wave phenomena hinder water depth estimation via optical radiative transfer in passive remote sensing, they form the foundation of bathymetric methods based on wave texture analysis. The synergistic use of active and passive imagery with wave textures helps to mitigate these adverse effects in certain inversion approaches. By integrating the inversion results from both data types—accounting for their distinct imaging mechanisms and depth detection theories—their complementary advantages can be effectively combined. To this end, a new fusion method based on a super learner–neural ODE (SL-NODE) ensemble is proposed. This two-stage framework implements a decision-level fusion strategy: multiple classical machine learning models act as base learners in the first stage, and the neural ordinary differential equation (Neural ODE) serves as the metalearner in the second. Outputs from the first-stage models are combined via cross validation and subsequently refined by the neural ODE to produce the final water depth estimates. In the range of 0–40-m depth range, the fused results exhibit reductions of over 17% in mean relative error (MRE) and more than 3 m in mean absolute error (MAE) compared to those from individual inversion methods. Moreover, the proposed ensemble achieves superior fusion accuracy to any single-classical machine learning model. The integration of multisource data and diverse inversion techniques significantly enhances the accuracy and utility of water depth retrieval from imagery with wave phenomena.**

***Index Terms*—Active and passive remote sensing images with waves, neural ordinary differential equation (Neural ODE), super learner, water depth inversion.**

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I. Introduction

THE water depth is crucial for human maritime activities. Traditional methods for in situ water depth measurements often pose challenges in inaccessible or unfavorable areas. Remote sensing emerges as a key technology for mapping extensive shallow-water zones, offering advantages in cost-effectiveness and high spatiotemporal resolution [1], [2]. Passive optical remote sensing imagery and synthetic aperture radar (SAR) represent the primary data sources for water depth retrieval. Among these, passive optical imagery provides an authentic visualization of the sea surface with strong interpretability and the ability to capture detailed marine features. However, it is prone to interference from atmospheric conditions—such as cloud cover and water vapor—which can lead to data gaps or degraded image quality. In scenarios requiring urgent response under adverse weather or in perennially cloudy and rainy regions, SAR data are often prioritized. Conversely, SAR does not directly reflect sea surface conditions. Each type of imagery possesses distinct strengths and limitations, leading to varied performance in water depth applications.

Waves in active microwave imagery have been utilized for water depth inversion [3], [4]. Theoretically, this approach can derive water depth without ground-based data; however, the accuracy and resolution of such wave-texture-based methods are generally limited. Similarly, wave signatures in passive optical imagery can also enable water depth estimation [5], [6], though these techniques are less widely adopted than those based on the radiative transfer theory. Methods rooted in optical radiative transfer theory, commonly applied to passive optical imagery [7], [8], achieve high accuracy and resolution but rely on high-quality ground-based data and are unsuitable for turbid waters [9]. Furthermore, water depth inversion based on the radiative transfer tends to favor images with a smooth sea surface, as wave patterns can obscure subsurface features and degrade retrieval accuracy. However, in many marine environments, ocean waves are a pervasive phenomenon present in most remote sensing images and cannot be overlooked. Sea-surface conditions without waves are not always available, and such wave-free images are relatively scarce. Therefore, it is imperative to investigate methods for deriving accurate water depth data from remote sensing imagery affected by wave activity.

Currently, the fusion of active and passive remote sensing for bathymetric inversion typically involves combining

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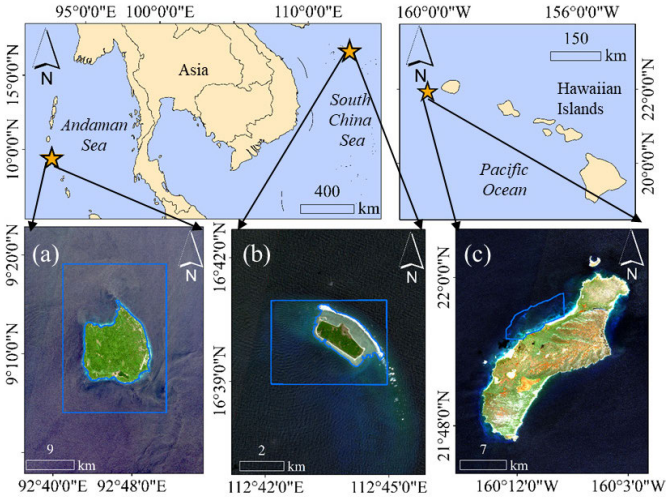


Fig. 1. Schematic map of the study area. (a) Car Nicobar Island. (b) Dongdao Island. (c) Niihau Island. The blue box indicates the specific study location.

spaceborne LiDAR data and passive remote sensing imagery to retrieve water depth information [10], [11]. [12], [13]. However, spaceborne LiDAR data are strongly influenced by sea surface waves, which introduce considerable noise and restrict the acquisition of bathymetric data in wave-impacted regions. This consequently limits their applicability in such areas. In wave-dominated waters, therefore, the availability of active and passive remote sensing imagery capable of capturing wave characteristics offers a new strategy for active–passive fusion in bathymetry. Currently, there are limited applications that collaboratively integrate active SAR and passive optical remote sensing imagery for water depth retrieval. Due to the distinct imaging mechanisms of the two modalities and the differing theoretical foundations of their respective water depth inversion methods, a feasible approach is to fuse the depth retrieval results derived from both active and passive remote sensing data. Both SAR and passive optical imagery can be processed using wave-based methods, whereas radiative transfer-based methods are exclusive to passive optical imagery. Among the latter, the log-linear method [14] and the Stumpf method [15] are widely adopted. Machine learning is a powerful approach for integrating diverse data types. Classical algorithms such as random forest, gradient boosting, and support vector regression (SVR) have demonstrated strong performance in remote sensing regression tasks [16], [17], [18], [19], [20], [21]. To complement their distinct predictive characteristics, the integration is performed within the super learner ensemble framework [22]. Furthermore, neural ordinary differential equation (Neural ODE) [23] is adopted as the metalearner for super learner to capture the complex nonlinear relationships among the base models' predictions. To address the need for fusing various water depth results from active and passive remote sensing, a super learner–neural ODE (SL-NODE) method is proposed to enhance the effectiveness of water depth inversion.

The remainder of this article is organized as follows. Section II describes the study area and data. Section III introduces various water depth inversion methods, followed by the proposed methodology in Section IV. Experimental results are presented in Section V, and a multidimensional discussion is provided in Section VI. Finally, the conclusion is given in Section VII.

II. Materials

A. Study Area

The study areas include Dongdao Island in the eastern Xisha Islands, Car Nicobar Island in the northern Nicobar Islands, and Niihau Island in the western Hawaiian Islands. Their geographical locations are shown in Fig. 1. Dongdao Island, administered by Sansha, Hainan, China, exhibits a rectangular shape and is surrounded by sandy banks. The island is formed from uplifted reefs and composite coral–shell sand deposits. Car Nicobar, the northernmost island of the Nicobar Archipelago in the Andaman Sea, is also rectangular, covering a total area of 126.91 km^2 with a maximum elevation of 60 m. Niihau, the seventh-largest island in U.S. Hawaiian chain, has an area of 179.9 km^2.

B. Data

The remote sensing imagery comprises active microwave (SAR) and passive optical data. SAR data were obtained from the Sentinel-1 mission, which operates in the C-band with a pixel resolution of 10 m. Optical data were sourced from Sentinel-2, with bands B2, B3, and B4 commonly employed for water depth inversion at a resolution of 10 m. The Sentinel-1 images for Dongdao Island, Car Nicobar Island, and Niihau were acquired on December 19, 2021, October 2, 2023, and November 12, 2024, respectively. Corresponding Sentinel-2 images were acquired on February 19, 2020, February 5, 2019, and November 22, 2022, respectively. Although the acquisition times of remote sensing images may differ, the temporal gap between SAR and optical imagery generally does not substantially affect the stability of the fusion results. On one hand, the derived water depth represents instantaneous depth, which is tide-corrected to obtain the water depth referenced to the theoretical datum; furthermore, the actual water depth in a given region is generally stable, so the time difference in image acquisition does not lead to significant accuracy degradation. On the other hand, errors associated with images acquired at different times are often partially independent. For independent or weakly correlated errors, the fusion of water depth estimates can effectively reduce random errors, yielding results that more closely approximate the true water depth.

Water depth data for Dongdao area were collected in situ using a shipborne single-beam sidescan sonar system, with depth and positional errors of 0.01 and 1 m, respectively. For the Car Nicobar area, water depth data were obtained from nautical charts compliant with the category of zone of confidence (CATZOC) standards established by the International Hydrographic Organization (IHO). The Niihau study area employed airborne LiDAR water depth data acquired by the scanning hydrographic operational airborne LiDAR survey (SHOALS) system. SHOALS is one of the most widely used airborne bathymetric LiDAR systems globally, capable of

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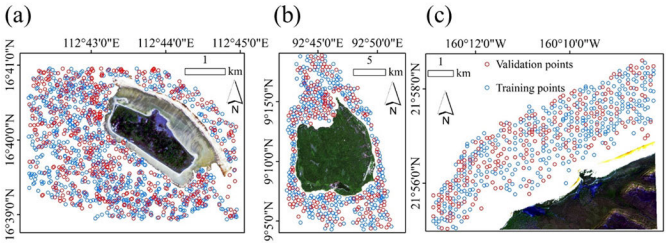


Fig. 2. Map showing locations of training and validation points in the study area. (a) Dongdao Island. (b) Car Nicobar Island. (c) Niihau Island.

achieving depths up to 40 m with a vertical error of less than 0.15 m. The locations of the in situ water depth points across the study areas are presented in Fig. 2. The numbers of training and validation points are 772 and 522 for Dongdao, 337 and 206 for Car Nicobar, and 334 and 221 for Niihau, respectively. The depth distribution of all points ranges from 0 to 40 m. The training and validation points in the three regions were split at a 3:2 ratio, which constitutes a reasonable and appropriately sized division. Spatially, both the training and validation points were uniformly distributed across the study areas, mitigating potential adverse effects caused by uneven spatial distribution.

III. Related Works

A. Water Depth Inversion Method Based on Wave Texture in SAR

The linear dispersion relation [24] describes the relationship between nearshore waves and the water depth h in shallow seas [25]: $\omega^2 = gk\tanh(kh)$. $\omega = 2\pi/T$ is the angular frequency ($\mathrm{rad}\cdot\mathrm{s}^{-1}$) and T is the wave period; g is the gravitational acceleration; $k = 2\pi/L$ is the angular wavenumber ($\mathrm{rad}\cdot\mathrm{m}^{-1}$) and L is the wavelength. The formula for obtaining water depth is as follows:

$$h = \frac{L}{2\pi}\operatorname{atanh}\left(\frac{2\pi L}{gT^2}\right). \tag{1}$$

Given that the wave period is largely conserved during propagation from deep to shallow water, the deep-water period can be used as a proxy for the shallow-water period [26]. The deep-water wavelength L_{deep}, which is stable due to minimal influence from underwater topography, can therefore be applied to calculate the deep-water period: $T_{deep} = (2\pi L/(g\tanh(2\pi h/L)))^{1/2}$ and $tanh(2\pi h/L) \approx 1$. Therefore, the period calculation formula can be simplified to: $T_{deep} = (2\pi L_{deep}/g)^{1/2}$. After applying the 2-D discrete fast Fourier transform (2-D FFT) to an image, a discrete frequency domain is obtained. The distance between two peaks in this frequency domain can be used to calculate the wavelength.

B. Water Depth Inversion Method Based on Wave Texture in Passive Optical Imagery

The principle of water depth inversion based on wave texture in optical imagery is the same as that in SAR imagery; both methods derive water depth from the empirical relationship between wave parameters and water depth. As waves propagate from deep to shallow coastal waters, their wavelength and velocity decrease with reducing water depth. Given the wavelength and wave velocity, the water depth can be obtained using the linear dispersion relation [27]

$$h = \frac{L}{2\pi}\operatorname{atanh}\left(\frac{2\pi c^2}{gL}\right) \tag{2}$$

where c is the phase speed. Wavelength L and phase φ can be derived from wave signals in two sequential images acquired with a very short time interval (a few seconds). Using the time interval Δt between the two images, the wave speed involved in the linear dispersion relation can be obtained from the wavelength and phase

$$c = \frac{Lp}{2\pi\Delta t}. \tag{3}$$

By applying FFT to the B2 band and B4 band images, respectively, the wave signal correlation F was obtained, and the phase φ was subsequently extracted from the significant wave information

$$\mathcal{F} = \mathcal{F}_1\mathcal{F}_2^* = Ame^{i\theta} \tag{4}$$

$$p = \operatorname{atanh}\left(\frac{k_y}{k_x}\right) \tag{5}$$

where $\mathcal{F}_1$ and $\mathcal{F}_2$ are the matrices derived from the FFT of the B2 band and B4 band images, respectively, Am represents the amplitude of the resulting spectrum [5], and k_x and k_y denote the spatial frequencies of the ocean waves in the x- and y-directions.

C. Water Depth Inversion Method Based on Radiative Transfer of Light

The log-linear and stumpf methods represent classical approaches for water depth inversion based on the optical radiative transfer theory. Building on the principles of subsurface light propagation, Lyzenga [14] proposed a multiband linear model for deriving water depth from multispectral imagery, expressed mathematically as follows:

$$\hat{Z} = \alpha_0 + \sum_{i=1}^{N}\alpha_i \ln\left[R_W(\lambda_i) - R_\infty(\lambda_i)\right] \tag{6}$$

where, $\hat{Z}$ is the model-derived water depth, $\alpha_i (i = 0, 1, \ldots, N)$ is the model parameters to be determined, N is the number of spectral bands, $R_W(\lambda_i)$ is the water-leaving radiance of band i, and $R_\infty(\lambda_i)$ represents the deep-water radiance of band i. This formulation, known as the log-linear model, is one of the most widely used methods for water depth inversion from multispectral remote sensing imagery.

Stumpf et al. [15] improved the log-linear method to prevent invalid results caused by higher deep-water radiance than shallow-water radiance, thereby extending its applicability to low-albedo regions. The modified expression is given as follows:

$$\hat{Z} = m_1 * \frac{\ln\left[nR_W(\lambda_i)\right]}{\ln\left[nR_W(\lambda_j)\right]} + m_0. \tag{7}$$

In the formula, m_1 is a scaling coefficient adjusted according to water depth, and n is a band-tuning factor that ensures the logarithm remains positive and physically meaningful. The

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terms $R_W(\lambda_i)$ and $R_W(\lambda_j)$ represent the apparent reflectance in bands i and j, respectively, while m_0 denotes the offset term for zero-meter water depth.

D. Water Depth Inversion Method Based on Radiative Transfer of Light

The accuracy of water depth inversion is typically quantified using the mean absolute error (MAE) and mean relative error (MRE). MAE measures the average magnitude of absolute differences between predicted and observed values, reflecting the mean of residuals. MRE represents the average of absolute relative errors, indicating the scale of relative deviation. For a given absolute error, the relative error decreases as water depth increases. Higher values of both MAE and MRE indicate greater error and lower precision in the derived water depth. These indexes are mathematically defined as follows:

$$MAE = \frac{\sum_{i=1}^{N} |Z_i - X_i|}{N} \tag{8}$$

$$MRE = \frac{\sum_{i=1}^{N} \frac{|Z_i - X_i|}{Z_i}}{N}. \tag{9}$$

In the above formula, Z_i denotes the measured water depth value at the ith point, X_i denotes the inverted water depth value at the ith point, and N denotes the total number of water depth measurement points.

IV. Fusion Method for Bathymetric Inversion Using Active and Passive Remote Sensing Imagery

A. Basic Fusion Methods

Stable and reliable machine learning methods serve as the foundation for bathymetric inversion and subsequent fusion-based depth retrieval. Given that random forest, gradient boosting, multilayer perceptron (MLP), elastic net regression, SVR, and K-nearest neighbors (KNNs) are classic, robust machine learning methods with validated performance across diverse fields, these six approaches were selected as base learners. The random forest is capable of handling regression tasks [16], [28], [29]. Each decision tree in the forest is built using a randomly sampled subset of the data, with individual trees trained on different portions of the complete dataset. The final prediction is generated by aggregating the results from all decision trees [30]. The gradient boosting regression tree algorithm [17] begins by defining an optimized loss function and initial values. It then constructs individual regression trees for prediction and iteratively adds new trees to minimize the loss function, ultimately forming an ensemble model [31]. MLP is a fundamental artificial neural network consisting of multiple layers of neurons, widely used for regression tasks in machine learning [18]. MLP employs the backpropagation algorithm for training: this algorithm calculates the error between predicted and actual outputs, propagates the error backward through the network, and updates the network parameters layer by layer to gradually converge toward the optimal solution. The elastic net regression is a linear regression approach that incorporates two regularization parameters. It enables effective feature selection on high-dimensional data while addressing multicollinearity among features [19]. SVR constructs nonlinear regression models using kernel functions and margin constraints. It aims to identify a regression hyperplane that minimizes the distance to the data points [32]. As a powerful regression model, SVR is suitable for handling nonlinear problems and scenarios with limited samples. KNN is a straightforward nonparametric algorithm applicable to regression problems [21]. Unlike methods that involve explicit model fitting, KNN predicts by identifying the K most similar neighbors from the training data.

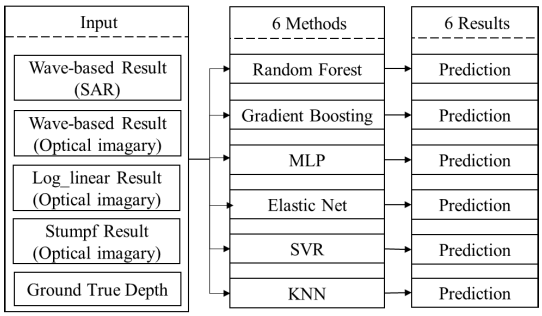


Fig. 3. Fusion flowchart of multiple classical machine learning methods.

In the experiments, the parameters for each machine learning method were configured as follows. For random forest, the number of decision trees, maximum tree depth, and random state were set to 100, 10, and 42, respectively. For gradient boosting, the number of decision trees and random state were set to 100 and 42, respectively. For MLP, the hidden layer structure was set to 200, and the maximum number of iterations was set to 100. For thw elastic network regression, the total regularization strength (alpha) was set to 0.1, and the L1 regularization ratio was also set to 0.1. For SVR, the regularization parameter was defined as [0.1, 1, 10], the insensitive bandwidth epsilon was defined as [0.2, 0.1], and the kernel coefficient gamma was defined as ["scale," 0.2, 1]. For KNN, the number of nearest neighbors was defined as [20, 30, 50, 100].

Machine learning methods integrate the advantages of both active and passive remote sensing for water depth inversion, screening out outliers with significant errors. Fig. 3 shows the fusion workflow based on classical machine learning. The input data include wave-based water depth results (from SAR and passive optical imagery), light radiation transfer-based results (from passive optical imagery), and in situ measurements. The machine learning model synthesizes these inputs to produce a final water depth dataset, yielding six distinct fused products.

B. Fusion Approach Based on SL-NODE

The super learner ensemble method is an advanced stacking algorithm that integrates predictions from multiple machine learning models through cross-validation and employs a meta-model to learn their optimal combination [22], [33], [34]. Neural ODEs [23], [35] formulate neural network dynamics using differential equations. Neural ODE is chosen as the meta-model due to its continuous dynamical characteristics, which improve generalization and help to mitigate overfitting. In contrast to simple linear weighting, Neural ODE captures

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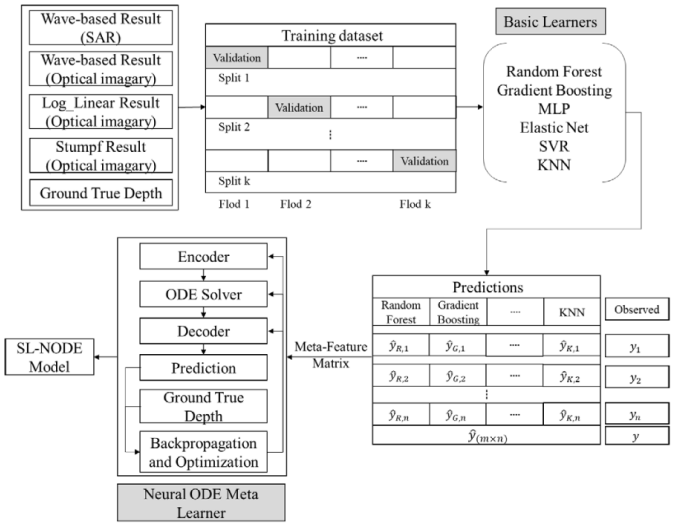


Fig. 4. SL-NODE method flowchart.

complex nonlinear relationships among water depth predictions from base models, thereby enabling the identification of deeper integration patterns. The core concept of Neural ODE lies in generalizing the discrete layer transformations in traditional neural networks into the evolution of a continuous dynamical system. Although water depth data used as input features do not inherently exhibit temporal sequential properties, Neural ODE encodes the input vector as an initial point in a continuous state space, describes the evolution trajectory of this state with respect to "pseudo-time" via a parameterized ordinary differential equation, and finally outputs the predicted value through a decoder. Traditional linear metamodels assume that the contributions of individual base models are independent and that their combination weights remain constant. However, in practical applications, the relative performance of different base models often varies across different input regions. Through its continuous evolution process, Neural ODE can learn input-dependent dynamic combination strategies in the state space; that is, it adaptively determines how to fuse information based on the vector distribution of predictions from the base models. Accordingly, the SL-NODE ensemble method is proposed to fuse multiple water depth inversion results. The overall architecture is depicted in Fig. 4.

The super learner ensemble method employs k-fold cross-validation for each base model, which enhances the prediction accuracy [33], [36]. The core idea of k-fold cross validation is to partition the water depth training dataset into k similarly sized subsets. Through k rounds of training and validation, more reliable performance estimates are obtained. For each base model, the training set is divided into k folds. In each iteration, k − 1 folds are used for training and the remaining one for testing, ensuring that each sample is included in the test set exactly once. This process yields an n-dimensional vector of water depth predictions for each base model, where n is the total number of training samples. These prediction vectors from multiple base models are then assembled into an n × m metafeature matrix, where m represents the number of base models. For medium-sized datasets, fivefold cross validation is commonly adopted, as it ensures sufficient data for training while providing reliable validation performance.

The generated water depth metafeature matrix serves as input, with the true target variable as output. The Neural ODE metamodel learns to optimally combine the predictions from the base models into a final prediction. An encoder (linear layer) first maps the metafeatures to a hidden state. This hidden state is then evolved through an ODE solver, which solves the ordinary differential equation governing its dynamics. Finally, a decoder (linear layer) maps the resulting hidden state to the final predicted value. The hidden layer dimension and learning rate are key parameters that govern the model capacity and training process of Neural ODE. The hidden layer dimension, which represents the number of neurons in the hidden layer of the internal network, was set to 32, meaning that 32 "feature extractors" are used to represent the ODE state of each sample. The learning rate, which controls the step size for weight updates during optimization, was set to 0.01. The core equation of the Neural ODE is as follows:

$$\frac{dh}{dt} = f(h(t), t, \theta) \tag{10}$$

where $h(t)$ denotes the hidden state at virtual time t, and θ represents the parameters of the ODE function—specifically, the weights and biases of neural network f, which defines the instantaneous rate of change of the hidden state. Although time t is treated as continuous in standard Neural ODEs, the metamodel in this study is applied to static data without explicit temporal sequences. In such cases, t serves as a dummy variable representing the evolution process rather than physical time. Notably, the proposed Neural ODE constitutes an autonomous system, meaning that the differential equation does not explicitly depend on time.

The mathematical representation of the numerical ODE solver is given by (11), which integrates the system from $t = 0$ to $t = T$ to obtain the final hidden state. For autonomous systems, these numerical methods remain applicable, as the introduction of virtual time t serves solely to parameterize the evolution trajectory of the hidden state

$$h(T) = h(0) + \int_0^T f(h(\tau); \theta) d\tau. \tag{11}$$

The ODE is solved using DOPRI5, the Dormand–Prince 5 (4) order adaptive step-size Runge–Kutta method. As an explicit method, it features error control and adaptive step-size adjustment [37]. The DOPRI5 scheme estimates the local truncation error by comparing two solutions of different orders, with h_{5th} and h_{4th} being the fifth-order and fourth-order solutions, respectively,

$$h_{5th} = h(t) + \Delta t \sum_{i=1}^{7} b_i k_i \tag{12}$$

$$h_{4th} = h(t) + \Delta t \sum_{i=1}^{7} \widehat{b}_i k_i. \tag{13}$$

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Here, b_i and $\widehat{b_i}$ represent the weight coefficients for the fifth-order and fourth-order solutions, respectively, while k_i denotes the seven intermediate gradient evaluations given by

$$k_1 = f(h(t);\theta) \tag{14}$$

$$k_2 = f(h(t) + \Delta t \cdot a_{21}k_1;\theta) \tag{15}$$

$$k_3 = f(h(t) + \Delta t \cdot (a_{31}k_1 + a_{32}k_2);\theta) \tag{16}$$

$$\cdots$$

$$k_7 = f\left(h(t) + \Delta t \cdot \sum_{j=1}^{6}(a_{7j}k_j);\theta\right). \tag{17}$$

The elements a_{ij} of the Runge–Kutta coefficient matrix are used to compute the slopes at intermediate steps. The adaptive step-size adjustment is determined by (18), where *tol* denotes the error tolerance

$$\Delta t_{new} = \Delta t\left(\frac{tol}{\|h_{5th} - h_{4th}\|}\right)^{1/5}. \tag{18}$$

The decoder is structured as a linear layer without an activation function, designed to predict a continuous water depth value, as given by

$$\hat{y} = W_{dec}h(T) + b_{dec} \tag{19}$$

where $W_{dec} \in \mathbb{R}^{1\times d_h}$, $b_{dec} \in \mathbb{R}$, and d_h denotes the hidden state dimension.

The backpropagation and optimization are essential components in neural network training, enabling models to automatically adjust their parameters through data to accomplish specific tasks. The loss function L is computed using mean squared error, with gradients calculated via backpropagation

$$\mathcal{L} = \frac{1}{m}\sum_{i=1}^{m}(y_i - \widehat{y_i})^2 \tag{20}$$

$$\nabla_\theta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h(T)} \cdot \frac{\partial h(T)}{\partial \theta}. \tag{21}$$

V. Experimental Results

A. Results of Basic Water Depth Inversion Method

The accuracy of three water depth inversion methods, SAR wave-based, passive optical wave-based, and light radiation transfer-based, was evaluated. The log-linear and Stumpf methods, two classical radiation transfer-based approaches, are commonly used as benchmarks in water depth inversion. Fig. 5 shows inversion results for Dongdao Island. The log-linear model achieved the highest accuracy, while the SAR wave-based method performed the poorest. Both radiation transfer-based methods produced negative values in some areas, a known limitation of such techniques. At Car Nicobar Island, the log-linear method significantly outperformed other approaches, with the optical wave-based method being the least accurate. Although the Stumpf method showed higher quantitative accuracy than wave-based methods, its outputs were systematically biased, with all estimates clustered between 20 and 30 m. In Niihau, the log-linear method remained most accurate despite some negative values. The optical wave-based method again showed the lowest accuracy and highest scatter, while the Stumpf method continued to produce nearly identical values. Overall, radiation transfer-based methods performed better, likely due to the incorporation of depth control points. Wave-based methods, which use no prior depth information, showed lower accuracy but avoided the negative values common in radiation transfer-based approaches.

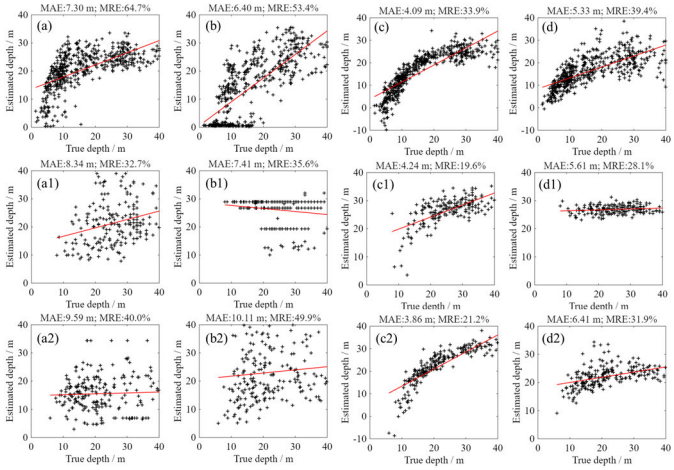


Fig. 5. Scatter plots of water depth inversion results. (a)–(d) SAR wave-based, optical wave-based, log-linear, and Stumpf method results from Dongdao Island respectively. (a1)–(d1) Car Nicobar Island. (a2)–(d2) Niihau Island.

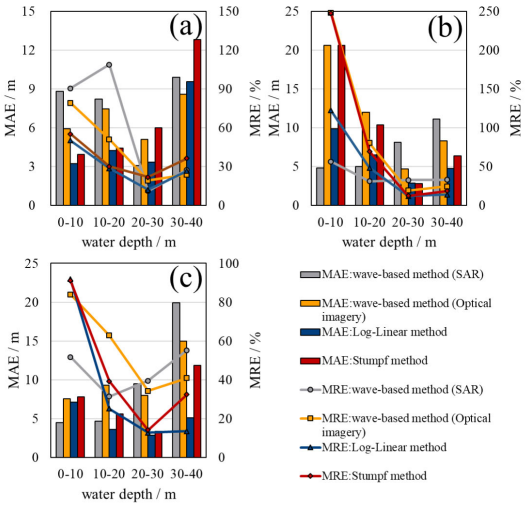


Fig. 6. (a)–(c) Segmented water depth accuracy maps for the four single-class water depth inversion methods on Dongda Island, Car Nicobar Island, and Niihau Island, respectively.

Water depth inversion accuracy varies across different depth ranges. Fig. 6 presents the segmented accuracy of depth results for the three study areas. In terms of MRE, the 0–10-m depth range shows the highest MRE, while the range of 30–40 m exhibits the lowest. The 10–20- and 20–30-m ranges demonstrate intermediate MRE values. This pattern occurs because smaller absolute errors produce larger relative errors in shallow waters. First, in the 0–10-m shallow water zone, sensor signal-to-noise ratio (SNR) limitations tend to amplify minor radiometric errors into relatively large depth errors. Second, bubble scattering induced by wave breaking

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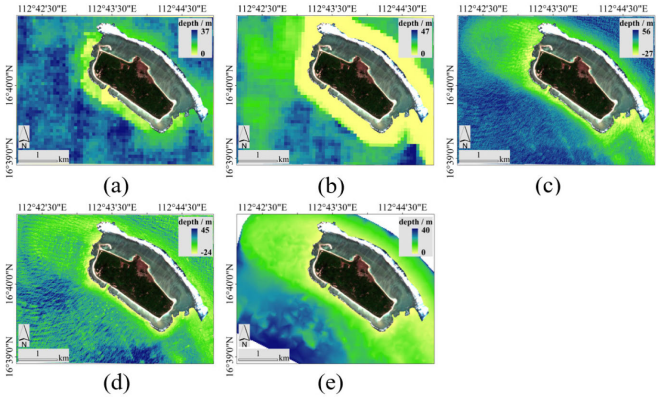


Fig. 7. Gridded results of single-class water depth inversion methods for the Dongdao Island. (a) SAR wave-based method. (b) Optical wave-based method. (c) Log-linear method. (d) Stumpf method. (e) Water depth DEM.

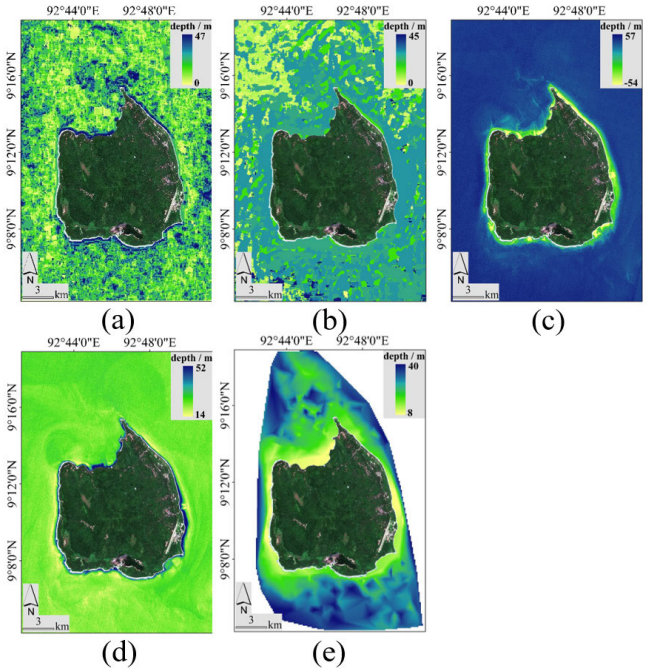


Fig. 8. Gridded results of single-class water depth inversion methods for the Car Nicobar Island. (a) SAR wave-based method. (b) Optical wave-based method. (c) Log-linear method. (d) Stumpf method. (e) Water depth DEM.

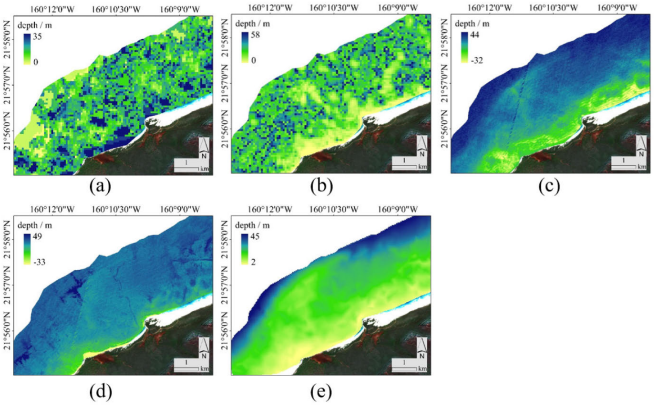


Fig. 9. Gridded results of single-class water depth inversion methods for Niihau Island. (a) SAR wave-based method. (b) Optical wave-based method. (c) Log-linear method (d) Stumpf method. (e) Water depth DEM.

effectively reduces the SNR, while increased interference from seabed spectral heterogeneity further complicates depth retrieval. Together, these factors account for the lower prediction accuracy observed in this depth range. Regarding MAE, the 30–40-m range shows the highest values in Dongdao and Niihau, whereas the 0–10-m range has the highest MAE in Car Nicobar. As shown in Fig. 6(a), for Dongdao Island: SAR wave-based inversion performs poorest in 0–10- and 10–20-m ranges; Stumpf method shows lowest accuracy in 20–30- and 30–40-m ranges; log-linear method achieves highest accuracy in 0–10, 10–20, and 20–30 m ranges; optical wave-based method performs best in the 30–40 m range. Fig. 6(b) reveals that for Car Nicobar Island: optical wave-based method and the Stumpf method perform poorly in 0–10- and 10–20-m ranges, while the SAR wave-based method shows best performance; in 20–30- and 30–40-m ranges, the SAR wave-based method performs worst, and the log-linear method performs best. According to Fig. 6(c), for Niihau Island: Stumpf method shows lowest accuracy in 0–10-m range, while SAR wave-based method performs best; in 10–20-m range, optical wave-based method is least accurate and log-linear method most accurate; in 20–30- and 30–40-m ranges, SAR wave-based method is least accurate while log-linear method performs best. The optimal method varies across different depth ranges and regions, demonstrating the necessity of integrating these four water depth inversion approaches.

Figs. 7–9 present the water depth raster results for Dongdao, Car Nicobar, and Niihau Islands, respectively. Wave-based inversion methods yield low-resolution results with prominent mosaic patterns, indicating inadequate characterization of seabed topography. Radiation transfer-based methods exhibit distinct wave artifacts and frequently produce negative depth values, with the Stumpf method particularly prone to generating homogenized depth estimates. Radiation transfer-based methods provide better representation of shallow water depth in Dongdao and Niihau, likely due to superior water penetration clarity in the passive optical imagery of these regions. In contrast, Car Nicobar's optical imagery shows minimal and uniform wave patterns, resulting in overly similar depth outputs from optical wave-based methods. SAR-based results in this area display pronounced mosaic artifacts due to inherent noise contamination. The inadequate recording of underwater topography in Car Nicobar's passive optical imagery further explains the poor performance of radiation transfer-based approaches in this region.

B. Fusion Results of Active and Passive Remote Sensing Water Depth Data

Given the complementary strengths and limitations of the four water depth inversion methods described above, decision-level fusion of their respective results is performed. The inversion outputs are integrated using multiple classical machine learning methods along with the proposed SL-NODE approach, with the subsequent accuracy assessment and

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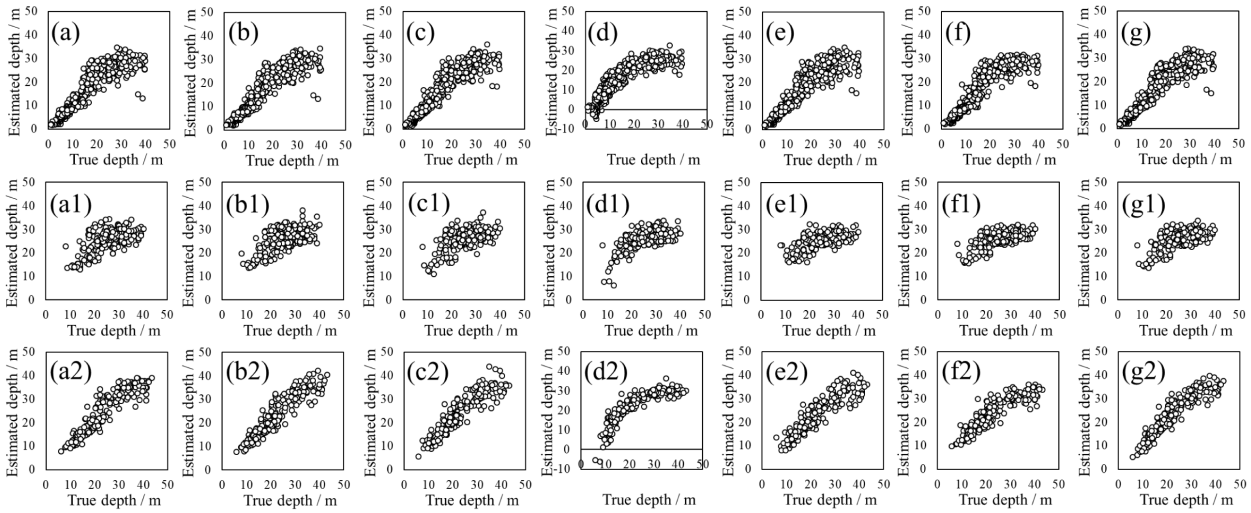


Fig. 10. Scatter plots of fused water depth results derived from active and passive remote sensing data using multiple fusion methods. (a)–(g) Random forest, gradient boosting, MLP, elastic net regression, SVR, KNN, and the proposed method in the Dongdao Island area. (a1)–(g1) Car Nicobar Island. (a2)–(g2) Niihau Island.

TABLE I

ACCURACY RESULTS FOR FUSION. BOLD BLACK INDICATES HIGHEST ACCURACY, BLUE INDICATES SECOND-HIGHEST ACCURACY, AND RED INDICATES LOWEST ACCURACY

METHOD	DONGDAO		Car Nicobar		Niihau	
	MAE (m)	MRE (%)	MAE (m)	MRE (%)	MAE (m)	MRE (%)
Random Forest	2.42	16.0	4.29	18.5	2.56	11.9
Gradient Boosting	2.43	16.4	4.34	18.9	2.53	11.8
MLP	2.47	16.1	4.19	**18.2**	2.63	12.2
Elastic Net	3.62	29.6	4.24	18.7	3.77	19.7
SVR	2.46	15.7	4.35	19.8	2.69	12.9
KNN	2.54	17.0	4.37	20.0	3.13	15.4
SL-NODE	**2.37**	**15.3**	**4.17**	18.3	**2.46**	**11.5**

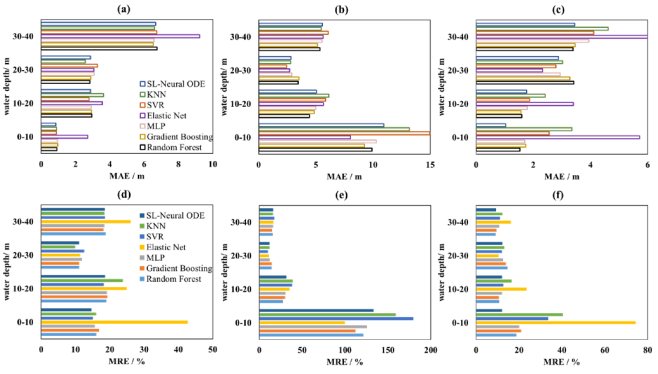


Fig. 11. (a)–(c) MAE of segmented water depths for active and passive water depth inversions obtained using various fusion methods on Dongdao Island, Car Nicobar Island, and Niihau Island, respectively. (d)–(f) MRE.

comparison of the fused products. Fig. 10 presents scatter plots of the fused results, showing that the SL-NODE method yields the least scattered distribution, indicating the superior accuracy. Furthermore, compared with the single-method results, the fused water depth points demonstrate noticeably tighter clustering, confirming the positive effect of data fusion on estimation accuracy. Integrating both image types and inversion methodologies helps to achieve more reliable water depth accuracy and higher spatial resolution, mitigating the limitations of single-source data and improving overall estimation robustness.

While scatter plots offer intuitive visualization, quantitative accuracy assessment is also required. Table I presents the accuracy metrics of various fusion methods across the three study regions. The SL-NODE method achieves the lowest errors in all areas. Specifically, in Dongdao Island, SL-NODE reduces MAE and MRE by 1.25 m and 14.3% compared with the poorest performing method; in Car Nicobar Island, by 0.2 m and 1.7%; and in Niihau Island, by 1.31 m and 8.2%. Regarding method ranking: random forest, MLP, and gradient boosting achieve the second-highest accuracy after the proposed method in Dongdao, Car Nicobar, and Niihau, respectively. The least accurate methods are elastic net, KNN, and elastic net across the three regions, while SVR demonstrates relatively stable performance without extreme results. The consistent superiority of SL-NODE can be attributed to its ensemble framework that integrates multiple classical machine learning methods, with the Neural ODE component further capturing complex nonlinear dynamics among their predictions.

Fig. 11 displays the segmented accuracy across the three study areas. Compared with Fig. 6, only a small number of fused water depth MRE values exceed 100%, appearing solely in the 0–10-m depth range in Fig. 11(e). Similarly, the MAE of fused results generally remains below 10 m. In Dongdao Island, the best-performing methods by depth range are: SL-NODE (0–10 m), SVR (10–20 m), KNN (20–30 m), and gradient boosting (30–40 m). The poorest performing methods are: elastic net (0–10 m), KNN (10–20 m), SVR (20–30 m), and elastic net (30–40 m). For Car Nicobar Island, the top methods are: elastic net (0–10 m), random forest (10–20 m), SVR (20–30 m), and gradient boosting (30–40 m). The weakest methods are: SVR (0–10 m), KNN (10–20 m), gradient boosting (20–30 m), and SVR (30–40 m). In Niihau Island, the leading methods are: SL-NODE (0–10 m),

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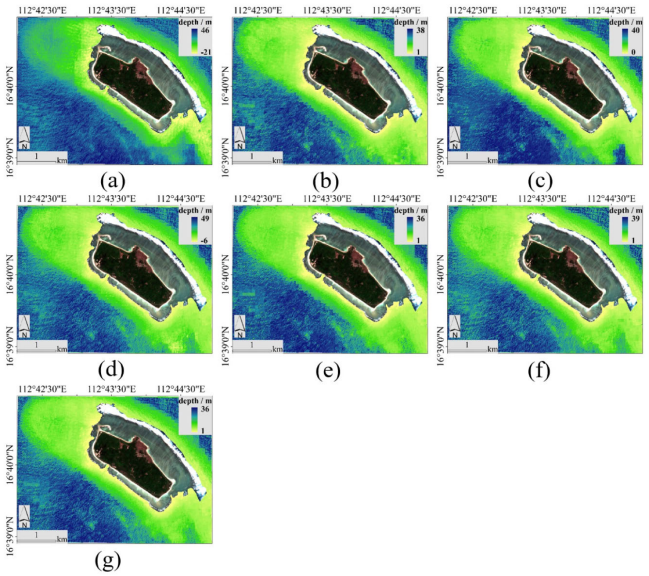


Fig. 12. (a)–(g) Water depth raster maps of Dongdao Island generated using random forest, gradient boosting, MLP, elastic net regression, SVR, KNN, and the proposed method, respectively.

gradient boosting (20–20 m), elastic net (20–30 m), and random forest (30–40 m). The least accurate methods are: elastic net (0–10 m, 10–20 m, and 30–40 m) and random forest (20–30 m). Notably, SL-NODE never ranks as the worst-performing method in any depth range across all study areas.

Fig. 12 presents the fused water depth raster for Dongdao Island. Compared with single-method inversions, the fused results exhibit reduced wave artifacts and more distinct shallow-deep water boundaries. However, negative depth values persist in results from random forest and elastic net. Fig. 13 shows the fused results for Car Nicobar Island. While the mosaic pattern remains largely unmitigated, water depth features around the island appear clearer. This limited improvement may be attributed to the near-uniform depth values derived from radiation transfer-based methods in this region, which fail to counterbalance the wave-induced mosaic artifacts. Among all methods, SL-NODE produces depth values that best align with realistic bathymetric patterns. Fig. 14 shows the fused results for Niihau Island. The fused results show no significant mosaic patterns or wave artifacts. SL-NODE yields particularly well-defined shallow-deep water boundaries and clearer coastal underwater topography, with minimal wave artifacts compared to other approaches. Overall, data fusion substantially enhances visual quality by mitigating the mosaic patterns typical of wave-based methods and the pronounced wave artifacts characteristic of radiation transfer-based approaches.

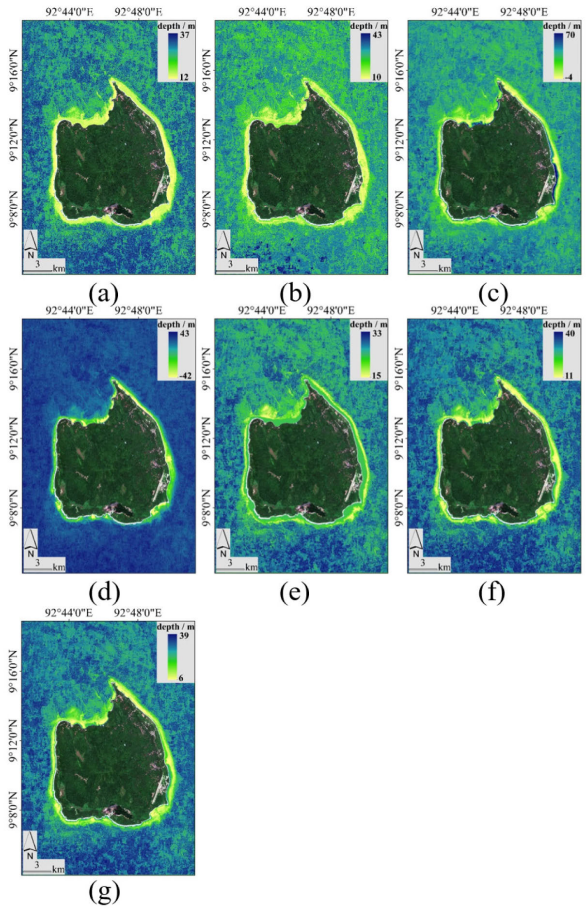


Fig. 13. (a)–(g) Water depth raster maps of Car Nicobar Island generated using random forest, gradient boosting, MLP, elastic net regression, SVR, KNN, and the proposed method, respectively.

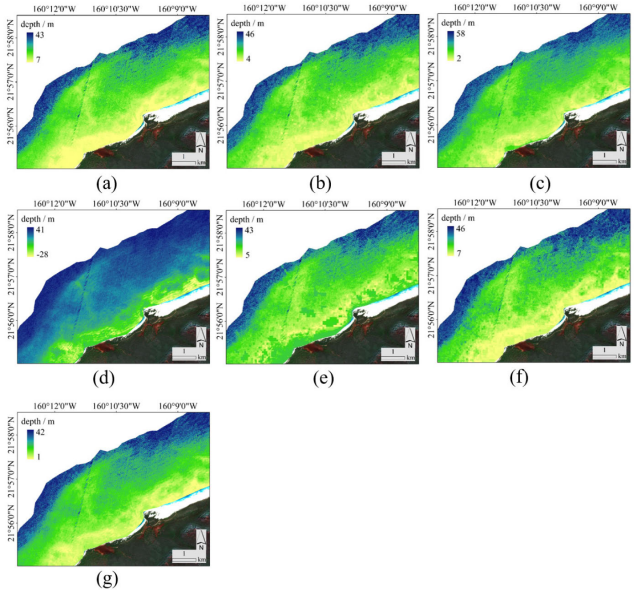


Fig. 14. (a)–(g) Water depth raster maps of Niihau Island generated using random forest, gradient boosting, MLP, elastic net regression, SVR, KNN, and the proposed method, respectively.

VI. Discussion

A. Fusion Analysis With Incomplete Active and Passive Water Depth Data

Given that certain types of data may be missing in practical applications, it is worthwhile to evaluate the fusion results under various data-missing scenarios and to retrain the SL-NODE model for each scenario.

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TABLE II

ACCURACY OF FUSION WITH MISSING SAR WAVE-BASED WATER DEPTH RESULTS. BOLD INDICATES THE HIGHEST PRECISION

METHOD	DONGDAO		Car Nicobar		Niihau	
	MAE (m)	MRE (%)	MAE (m)	MRE (%)	MAE (m)	MRE (%)
Random Forest	2.53	17.0	4.41	18.9	2.66	12.1
Gradient Boosting	2.50	16.8	4.31	18.8	2.59	11.8
MLP	2.53	16.6	4.18	18.5	2.58	11.9
Elastic Net	3.66	30.4	4.19	18.8	3.77	19.7
SVR	2.50	16.1	4.36	19.7	**2.52**	**11.6**
KNN	2.61	17.4	4.42	20.4	2.82	13.8
SL-NODE	**2.49**	**16.4**	**4.15**	**18.3**	2.56	11.7

TABLE III

ACCURACY OF FUSION WITH MISSING PASSIVE OPTICAL WAVE-BASED WATER DEPTH RESULTS

METHOD	DONGDAO		Car Nicobar		Niihau	
	MAE (m)	MRE (%)	MAE (m)	MRE (%)	MAE (m)	MRE (%)
Random Forest	2.83	17.9	4.31	18.7	2.61	12.0
Gradient Boosting	2.75	17.8	4.40	19.2	**2.48**	**11.5**
MLP	2.70	17.4	4.17	**18.0**	2.62	12.1
Elastic Net	3.99	33.2	4.24	18.6	3.74	19.6
SVR	**2.66**	**16.7**	4.28	19.4	2.59	12.2
KNN	2.79	18.1	4.32	20.1	2.77	13.2
SL-NODE	2.71	17.2	**4.15**	18.2	**2.48**	**11.5**

TABLE IV

ACCURACY OF FUSION WITH MISSING ACTIVE AND PASSIVE OPTICAL WAVE-BASED WATER DEPTH RESULTS

METHOD	DONGDAO		Car Nicobar		Niihau	
	MAE (m)	MRE (%)	MAE (m)	MRE (%)	MAE (m)	MRE (%)
Random Forest	3.02	19.4	4.53	19.6	2.73	12.3
Gradient Boosting	2.92	18.8	4.44	19.3	2.63	11.9
MLP	2.79	18	**4.11**	**18.3**	2.54	11.5
Elastic Net	3.97	32.3	4.19	18.8	3.74	19.6
SVR	**2.71**	**17.0**	4.31	19.5	**2.42**	11.3
KNN	2.84	18.5	4.26	19.5	2.59	12.3
SL-NODE	2.80	18.0	4.21	18.7	2.48	**11.2**

1) Fusion With Missing SAR Wave-Based Water Depth Data: The fusion accuracy under missing SAR wave-based water depth data is summarized in Table II. As shown in the table, the SL-NODE method achieves the best performance in Dongdao and Car Nicobar, while SVR performs best in Niihau. The absence of SAR wave-based results leads to decreased accuracy in Dongdao and partially reduced accuracy in Car Nicobar and Niihau. Both random forest and SL-NODE exhibit accuracy degradation across all three regions. These results indicate that missing SAR wave-based water depth data has an adverse effect on inversion performance.

2) Fusion With Missing Passive Optical Wave-Based Water Depth Data: Table III summarizes the fusion accuracy in the absence of passive optical wave-based water depth data. The absence of such data leads to performance degradation in Dongdao Island and partial accuracy reduction in the other two regions. SL-NODE achieves the best performance among all methods. In certain regions, missing passive optical wave-based water depth data can adversely affect the inversion accuracy to some extent.

3) Fusion With Missing Active and Passive Wave-Based Water Depth Data: The fusion accuracy under missing active and passive wave-based water depth data is presented in Table IV. In the absence of such data, Dongdao Island exhibits decreased accuracy across all fusion methods, while Car Nicobar and Niihau Islands show partial performance degradation with certain approaches. Under these conditions, SVR and SL-NODE achieve the superior performance in Dongdao and Niihau, whereas MLP and SL-NODE perform best in Car Nicobar. The observed accuracy reduction suggests that missing active and passive wave-based data adversely affect water depth inversion to some extent. This may be attributed to the complementary numerical characteristics provided by different inversion methods, which machine learning algorithms can effectively integrate to produce optimized fusion results.

TABLE V

FUSION ACCURACY WITHOUT RADIATION TRANSFER-BASED WATER DEPTH DATA

METHOD	DONGDAO		Car Nicobar		Niihau	
	MAE (m)	MRE (%)	MAE (m)	MRE (%)	MAE (m)	MRE (%)
Random Forest	**3.87**	**29.5**	5.71	26.7	6.89	37.1
Gradient Boosting	4.06	32.2	**5.19**	**24.8**	**6.81**	**36.7**
MLP	4.45	34.9	5.38	25.4	7.10	36.7
Elastic Net	5.06	40.7	5.55	26.5	7.24	37.4
SVR	4.35	32.8	5.47	26.3	7.17	33.1
KNN	3.93	30.3	5.44	25.8	7.20	37.3
SL-NODE	3.90	30.1	5.35	25.5	6.87	36.2

4) Fusion With Missing Radiation Transfer-Based Water Depth Data: The fusion accuracy under missing radiation transfer-based water depth data is presented in Table V. As shown in the table, all three regions exhibit accuracy degradation under these conditions. In Dongdao Island, random forest and SL-NODE demonstrate superior performance, while gradient boosting and SL-NODE achieve the best results in both Car Nicobar and Niihau Islands. The absence of radiation transfer-based data adversely affects fusion performance, likely due to the loss of high-accuracy characteristics inherent in this inversion approach. These results confirm the importance of incorporating radiation transfer-based methods to maintain optimal water depth fusion quality.

Radiation transfer-based water depth inversion includes the Log-linear and Stumpf methods, which share the same theoretical foundation and input data yet yield distinct results. Therefore, it is valuable to analyze scenarios involving the absence of either method's data separately. Fig. 15 presents heatmaps of MAE and MRE for fusion results under missing data from each method. The results demonstrate that fusion accuracy is significantly higher when Stumpf method data are omitted compared to cases missing Log-linear method data. This suggests that the log-linear method exhibits stronger water depth inversion capability in the presence of wave features in optical remote sensing imagery.

B. Application of Empirical-Statistical Water Depth Inversion Methods to Optical Imagery With Wave Textures

Empirical-statistical water depth inversion methods are widely used in optical remote sensing, though wave signals

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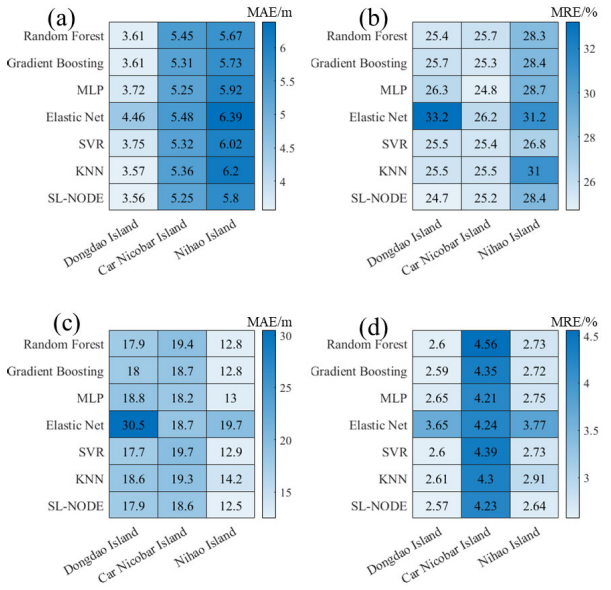


Fig. 15. (a) and (b) MAE and MRE heatmaps, respectively, of the seven fusion methods across the three regions when lacking log-linear method water depth results. (c) and (d) Corresponding heatmaps for cases missing Stumpf method water depth data.

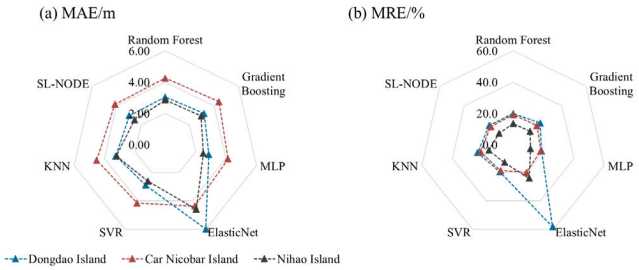


Fig. 16. (a) and (b) MAE and MRE, respectively, of water depth data derived from seven machine learning methods trained to directly model the relationship between pixel values and depth across the three regions.

can negatively impact their accuracy by interfering with pixel values that convey underwater information. Fig. 16 presents the accuracy of machine learning models trained directly on image pixel values across the three regions. Elastic net consistently performs the worst, likely because the high correlation among the red, green, and blue bands leads to suboptimal regression solutions. Multisource fusion generally outperforms these empirical methods. For example, in Dongdao, Car Nicobar, and Niihau regions, the MRE of SL-NODE with multisource fusion is 4.3%, 0.1%, and 0.1% lower than that of its empirical counterpart, while random forest fusion shows MRE reductions of 3.8%, 0.6%, and 1.6% under the same comparison.

C. Choice of the Metamodel

The metamodel constitutes the second layer of the super learner framework, taking as input the predictions from all first-layer base learners and generating the final ensemble output. The choice of meta-model is critical to the super

TABLE VI

WATER DEPTH INVERSION ACCURACY OF SUPER LEARNER WITH DIFFERENT METAMODELS. BOLD INDICATES THE HIGHEST PRECISION

METHOD	DONGDAO		Car Nicobar		Niihau	
	MAE (m)	MRE (%)	MAE (m)	MRE (%)	MAE (m)	MRE (%)
Random Forest	2.53	16.5	4.32	19.2	2.77	12.9
Gradient Boosting	2.79	17.8	4.75	20.8	3.12	14.1
NODE	**2.37**	**15.7**	**4.17**	**18.3**	**2.46**	**11.5**

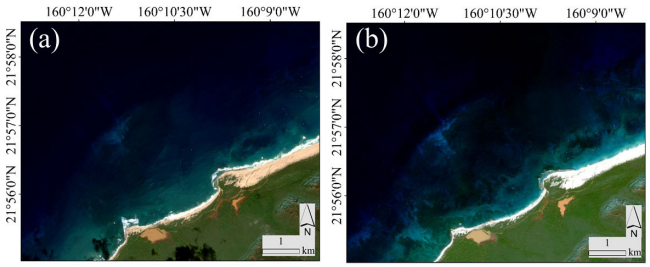


Fig. 17. Two images (a) and (b) were acquired on December 10, 2019 (denoted by I_1) and January 28, 2022 (denoted by I_2), respectively.

TABLE VII

ACCURACY OF WATER DEPTH INVERSION FROM IMAGES WITH DIFFERENT WAVE TRACE CONDITIONS

METHOD	I_0		I_1		I_2	
	MAE (m)	MRE (%)	MAE (m)	MRE (%)	MAE (m)	MRE (%)
Stumpf	6.41	31.9	6.07	29.4	5.43	25.6
Log_linear	3.86	21.2	3.85	20.5	3.06	14.4

learner's performance. Random forest and gradient boosting, both recognized as strong performers in water depth prediction, were selected as metamodels for comparison against the proposed Neural ODE-based metamodel. This comparison aims to evaluate the impact of metamodel selection on fusion accuracy. As shown in Table VI, the super learner with Neural ODE as metamodel achieves higher water depth inversion accuracy than those using random forest or gradient boosting. Furthermore, the random forest metamodel outperforms the gradient boosting variant. Therefore, Neural ODE is recommended as the metamodel for water depth data fusion.

D. Effects of Wave Texture on Radiative Transfer-Based Bathymetric Inversion Methods

For radiative-transfer-based bathymetric inversion methods, whether wave traces in optical imagery affect inversion accuracy is a question worth investigating. Given that the study areas are wave-prone regions, available images free of wave traces and cloud cover are relatively limited. In particular, no suitable Sentinel-2 imagery was found for Dong Island and Car Nicobar Island. Accordingly, this section uses Sentinel-2 imagery from Niihau Island that exhibits no wave traces for analysis. The two comparative images were acquired on December 10, 2019 (I_1) and January 28, 2022 (I_2), as shown in Fig. 17. The image employed in the preceding experiments is denoted as I_0.

Table VII summarizes the accuracy of water depth retrieval results from the three images obtained using the radiative

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transfer method. The inversion accuracy of the image without wave traces is higher than that of the images with wave traces. For example, the MRE of I_2 is 6.8% lower than that of I_0. This indicates that wave traces in remote sensing images exert adverse effects on radiative-transfer-based bathymetric inversion. The underlying reason is that wave traces violate the flat water surface assumption and introduce aliasing in the radiometric signal, resulting in retrieval errors.

VII. Conclusion

A considerable number of active and passive remote sensing images with wave signals are available in most marine areas. Improving the accuracy of water depth inversion from such imagery is of great importance. To address the limited accuracy of single-method inversion in wave-affected imagery, SL-NODE was adopted to integrate water depth results from both active and passive data, yielding a final fused depth product. The fused results exhibit superior accuracy and visual quality compared with those obtained by any single-inversion method. In some regions, using only one type of remote sensing data yields lower accuracy than jointly using both active and passive imagery, highlighting the practical value of fusing multisource water depth outputs. The integration of active and passive water depth data via machine learning offers three key benefits: it combines the complementary advantages of different sensors, reducing biases inherent to single-source data; it enhances the usability of wave-affected imagery; and it improves the overall accuracy of derived water depth.

The accuracy of the proposed method still has potential for improvement. Future work could explore incorporating multidimensional features or directly learning wave characteristics from ocean surfaces in active and passive imagery to achieve higher precision. In addition, spaceborne LIDAR offers an alternative source of water depth information. Its integration could enable water depth inversion without in situ measurements, though current spaceborne LIDAR is generally limited to clear shallow waters and requires high water quality.

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